# Homework 1

September 3, 2020

Herein, I will report my initial reactions to reading *Lectures in Advanced Mathematics: Why Students Might Not Understand What the Mathematics Professor Is Trying to Convey*

## 0.1 Responses

1. People absorb information in all sorts of ways. We test students' knowledge by asking them to solve problems which hopefully require them to recall what they "have been taught" and apply what they have learned. I think lecturing may be ineffective because the students are not really doing anything; sure, there are studies that say taking notes is doing something, and thus note-takers tend to learn more effectively, but I think that's a low bar for engagement as is reading (strictly speaking) a body of text or watching a video.

   I believe books offer an array of definitions and unmotivated concepts which students learn to understand and appreciate by working on exercises (e.g. proving a theorem for themselves or calculating something they have a frame of reference for). It's not obvious how to work through these exercises, which is why books often have long sections of preparatory material beforehand. Roughly, I believe an effective teacher of mathematics teaches their students how to read books (and other bodies of information), in a way that is fundamentally different from effectively reading it to them line by line. I think books and teachers should motivate the material so students understand why they are learning it, and we should expect students to lead their own learning process using teachers as resources to complement their books. Tldr: I don't think you can effectively teach a student anything they don't want to learn in any way; an effective teaching method begins by inspiring students to learn the material in the first place.

2. I think it's tragic. I think a mathematics education fundamentally prepares students for two things: (1) informal mathematical reasoning which they will apply throughout their lives, and (2) crystal-clear thought to guide or supplement that informal reasoning. I think teachers often get bogged down in ingraining crystal-clear thought into students, and miss out on opportunities to introduce and solidify foreign concepts which matter more. That is, I value the concepts and colloquial understanding of *small*ness and *toolbox* much more than the arithmetic of the proofs in a real analysis class. Rigorous proofs exist to solidify our understanding of smallness. I believe this is why students come to adopt the view that proofs are a set of steps rather than a methodology of problem solving.

3. After reading this article, I have come to value informal reasoning as the essence which I aim to communicate in my teaching, tutoring, and grading. I believe formal reasoning and calculation are stepping stones which aid in their development of informal reasoning, which I also value but to a lesser extent. The fundamental objectives in a first course on calculus, for example, are a deep informal understanding of limits, differentiation, integration, and an ability to identify problems which model them. The secondary objective is to be able to solve them for the kinds of functions we give to them. If this sounds counter-intuitive consider the strengths of an expert mathematician/educator. Would they be better able to explain what a limit is, or be able to compute it for some class of functions[1]?

[1] I know many great teachers which often "struggle" with these computations, and none that lack a fundamental understanding of the concepts in question.